Question 1: Historical Trends of UK Electrical Power

1.1 Historic trend analysis (2013-2024)

Seeing as new data from 2022-2024 is now available, it will be used to provide a more comprehensive overview of electricity from 2013-2024 and better predict future trends. Seen below are the net electricity generation and percentage breakdown of the UK from 2013-2024:

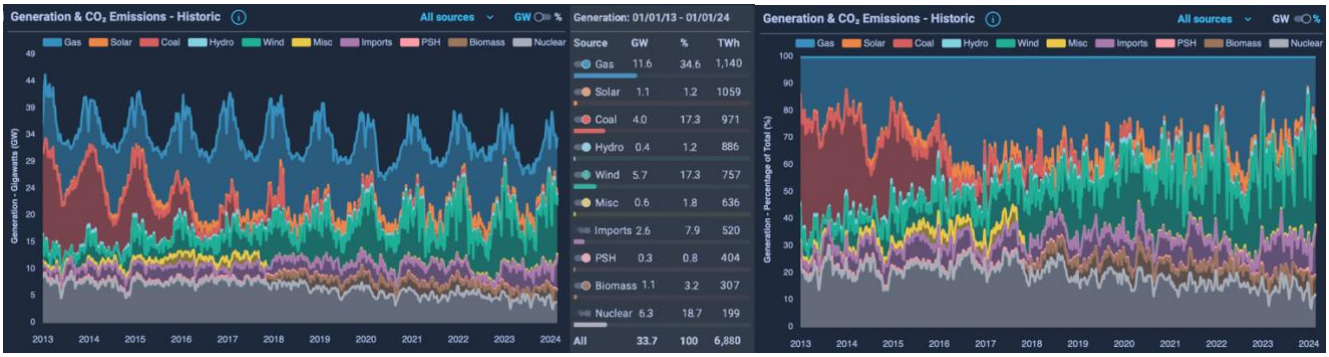


Fig 1 (Left) and 2 (Right): Electricity generation mix in terms of GW (left) and percentage breakdown (right). Adapted from [1], data is sourced from [2].

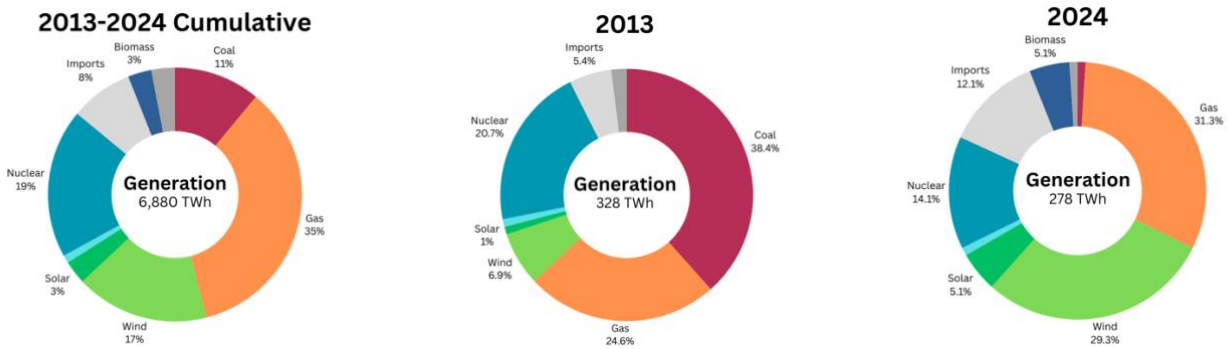


Fig 3a (Left), 3b (Middle) and 3c (Right): Electricity mix extracted from data in figure 1 and visualised via Pie chart.

From figure 1, it can be seen that the net generation of electricity has decreased. A net production of 328 TWh in 2013 is seen, and has since reduced and maintained at around 280-290 TWh across 2020-2024. From figures 2 and 3, a number of trends can be identified:

- The percentage contribution of fossil fuels has reduced greatly, from 63% of all generation in 2013 to 32.3% in 2024. This is due to the successful phase-out of coal power generation in the UK, reducing in contribution from 38.4% in 2013 to 1% in 2024. From 2016-2020, coal electricity production reached zero on many occasions, however production was restarted in August 2021 due to lack of wind [3][4] at the time.
- Despite the elimination of coal production, gas has remained a significant source of electricity, from 24.6% to 31.3% contribution from 2013-2024. This is understandable as it is considered a "transition fuel" as it produces fewer emissions than coal, meaning it is to be phased out afterwards. In addition, the UK has substantial existing infrastructure for natural gas, producing half of its own consumption as well as 85% of the housing stock relying on gas [5][6]. However, there are plans to phase out natural gas by 80% before 2035 [6].
- Electricity from renewable sources has increased greatly, spearheaded by the massive increase of wind power, from 6.9% to 29.3% across 2013-2024. This is not surprising due to regulations and demand favoring renewables, as well as the UK being among the best locations for wind power in not only Europe but the entire world [7][8]. Public opinion heavily favours wind power, with almost 75% of the UK agreeing to its use, including those living near onshore turbines [9]. Solar production has also increased, although far less substantially from 1% to 5.1%. Although the increase was fueled by cost reductions in photovoltaics as well as government tariffs [9], the overall solar potential of the UK is relatively low due to its high latitude [11].
- Imports of electricity have increased from 5.4% to 12.1%, and Nuclear power has reduced in its role from 20.7% to 14.1%. Hydroelectricity generation has stagnated at a 1% contribution due to low committed resources,

although pumped storage hydroelectricity (PSH) is being used in 4 sites, providing 1% of yearly output via storage [12]. Biomass has also emerged as a source from 2018 onwards, increasing to 5.1% by 2024.

1.2 Renewables capable of dislodging Fossil Fuels for 2022-2050

It is without a doubt that wind power is the most promising renewable source in the UK going forward. As previously mentioned, the UK is among the best locations in the world for wind power, and receives strong support from the public [7][9]. Across 2013-2024, production increased from 22 TWh to 81 TWh, making the UK the 6th largest producer of wind power in the world, and overtaking its own natural gas electricity production in Q1 2023 [13][14]. Currently, the UK has over 11,000 turbines producing a capacity of 30 GW, with 15 GW onshore and and 15 GW offshore [15]. The Government has also committed to an increase of 50 GW capacity by 2030, of which 5 GW will be floating offshore wind. Approval times for proposed offshore wind farms will also be reduced from 4 years to 1 year [16].

In addition, the Government currently issues Renewables Obligations Certificates (ROCs) to renewable energy generators for produced megawatt-hours of electricity, which can then be sold to electricity suppliers to demonstrate compliance with their obligations [17]. Although ROCs are intended for all renewable projects, wind energy benefitted substantially. Of the £6.3bn invested in ROCs in 2019-2020, 67% went into wind power [18]. A Contracts-For-Difference (CFD) scheme is also in place, providing long-term contracts to renewable energy generators that guarantee a set price for electricity produced. This provides revenue certainty for developers, attracting growth in wind projects [19]. Under the CFD scheme, £1.7bn was raised for wind power in 2020 [20]. With such promising demand, investment and locations, it is logical to assume wind energy will continue to spearhead renewable electricity production in the UK through to 2050.

Although solar power has a small role in UK electricity production at 5.1%. As previously mentioned, the UK has a relatively low solar potential due to its latitude, which is further hindered by its often overcast weather. However, one unexpected strength is that the UK's high winds can cool photovoltaic modules, increasing efficiency [21]. Government programmes are also in place, including the Smart Export Guarantee (SEG) implemented in 2020. This rewards homeowners or other small-scale electricity generators with payments for surplus electricity they export to the grid [22]. Although the aforementioned CFD scheme had originally disallowed solar PV projects from bidding since 2015, the Government has since reversed this decision in 2020. The main benefits are yet to be seen. Going forward, the costs of solar PVs are likely to further decrease due to advances in manufacturing and economies of scale. In addition, demand for small-scale installations will likely increase to meet demand from electric vehicles as well as increase energy security of homes [23]. Considering these factors, solar energy may not offset use of fossil fuels until phase-outs are complete, but will certainly become a notable contribute to UK energy.

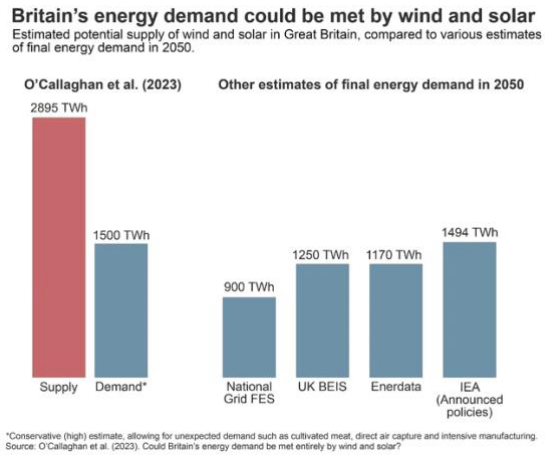
Question 2: Predictions for Electricity demand and mix 2023-2050

The global goal of reaching net zero carbon emissions by 2050 will likely result in increased electricity consumption per capita over time. Natural gas use is not only prevalent in electricity production, but also used in 85% of UK households for day-to-day activities such as cooking and heating [6][24]. A typical UK household consumes around 12-17 MWh of natural gas per year for heating and cooking [25]. For an estimated 28.2 million households in the UK, and using a middle-band average of 14.5 MWh/year per household, the net gas consumption equates to roughly 408.9 TWh of gas energy per year. For reference, the net production of electricity in the UK was 278 TWh, as seen in figure 3c. In addition, of the 278 TWh produced, 31.3% was from natural gas, equating to 87 TWh. Thus, as natural gas is phased out, an equivalent of 495.9 TWh of energy per year must be made up for through electricity from renewable energy. This does not include the increase in overall electricity demand per capita due to population growth, urbanisation and electrification of transport.

However, net demand for electricity may not necessarily trend upwards over time as continued improvements in energy-efficient standards, technologies and consumer behaviour are made [27]. This is reflected in figure 1, where net electricity production trends downwards from 2013-2020, and has stagnated at roughly 280 TWh per year. In addition, demand is also reflected by pricing. In 2022, due to soaring energy prices, domestic electricity consumption

fell by 10%, while industrial consumption was down by 2.7% and commercial users by 0.7% [28]. Considering these factors, the National Grid estimates that electricity consumption in the UK will increase by 50% by 2036 and more than double by 2050, to a value of over 560 TWh [24]. This will likely be accompanied by the phase-out of coal by 2024, and the phase-out of 80% of UK's natural gas by 2035 [29][30]. No additional goals are set on natural gas, other than the obligation to reach net zero carbon emissions by 2050 [31].

O'Callaghan et al. suggested in 2023 that the theoretical generating capacity of wind and solar power is 2,895 TWh by 2050, enough to meet the estimated electricity demand of 560 TWh as well as various predicted final energy demands of around 1200 TWh for 2050 [32][33]. This can be seen in figure 4 below. The majority of which will be generated by offshore wind, which will spread to roughly 10% of the UK's exclusive economic zone. Onshore wind would be used on 5% of land, and 2% of land used for solar PV, mostly on farmlands using a technique called agrivoltaics [33].



Source	Energy supply (TWh)	Land use
Utility-scale solar PV	544	2% of British land (Can be combined with farming)
Rooftop solar	25	8% of British rooftops
Onshore wind	206	5% of British land (Can be combined with farming)
Offshore wind (floating)	1557	8% of UK's exclusive economic zone
Offshore wind (fixed)	563	2% of UK's exclusive economic zone
Total	2896	

Fig 4 (Left): Comparison of predicted wind, solar energy production and final energy demand in 2050 [32].
Table 1 (Right): Prediction of energy generation mix for 2050 [32].

This prediction aligns with data seen in question 1, where the theoretical capacity of offshore wind for the UK is significantly higher than other renewables due to its long shorelines and high sea winds. The development of floating wind turbines will certainly accelerate the expansion of wind as the majority of the UK's exclusive economic zone has depths below 70m, the current upper limit for fixed-base offshore wind turbines. This is followed by solar PV production, where, as expected from analysis in question 1, a lesser portion of energy is produced. The UK has suboptimal conditions due to its weather and latitude, but due to cost reductions from technological developments as well as Government support, solar PVs have become more widely available [22][23]. This trend may very well continue towards 2050, leading to cheaper and more efficient solar PVs producing ample energy for homes and farms, reducing strain on the energy grid.

Question 3: Conceptual Design of Renewable Energy Source

3.1 Site location

Offshore wind will be considered for this question. Although onshore wind has historically been a cheaper option, offshore wind construction prices have fallen by one-third since 2012 due to improved technologies and support from the Government. In addition, offshore wind mean speeds tend to be much higher and consistent than onshore, providing higher availability and generation. This can be seen in the figures below, where from Figure 5b, only a fraction of onshore wind reaches the 10m/s rates seen on the oceans.

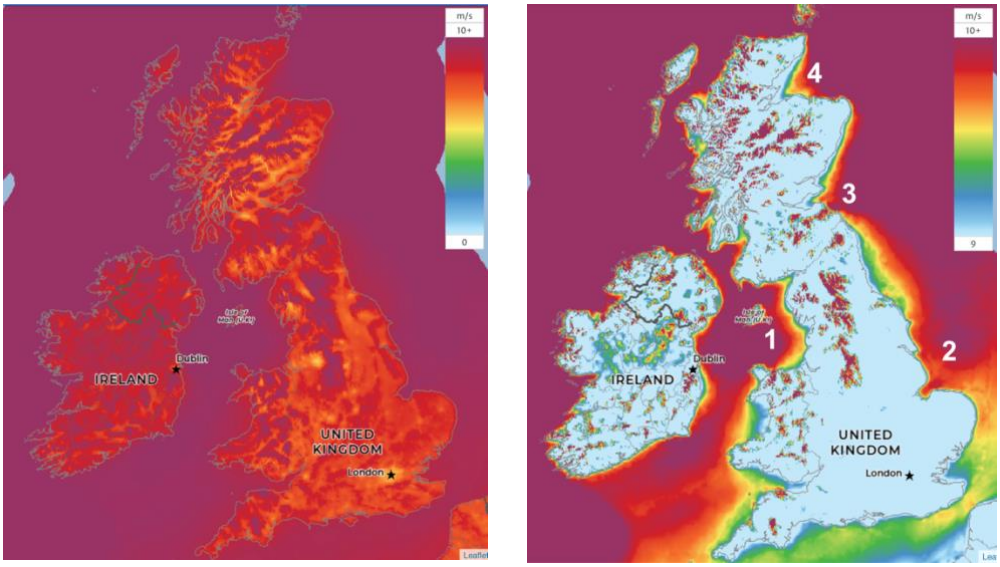


Fig 5a (Left) and Fig 5b (Right): Map of wind-speeds across the UK economic zone scaled from 0-10m/s (left) and 9-10m/s (right) [35].

The UK's long shoreline also provides immense resource availability of offshore wind, estimated at one-third of Europe's total offshore wind resources [36][37]. A study estimated that harnessing one-third of the UK shoreline 25m or shallower with wind turbines can produce up to 40 GW. For comparison, the average net UK generation from February 2023-2024 is 37 GW [38]. In addition, onshore wind farms require much space, and must compete with other uses such as agriculture or conservation. This prioritisation is also reflected by the Government, which has increased subsidies for offshore wind as of 2023 [39]. Again referring to figure 5b, it was determined that there were 4 prime locations for wind by considering wind speed and proximity to land. This is important to consider to optimise logistics and transmission requirements. This choice is reflected in the map of existing and planned offshore wind farms seen below:

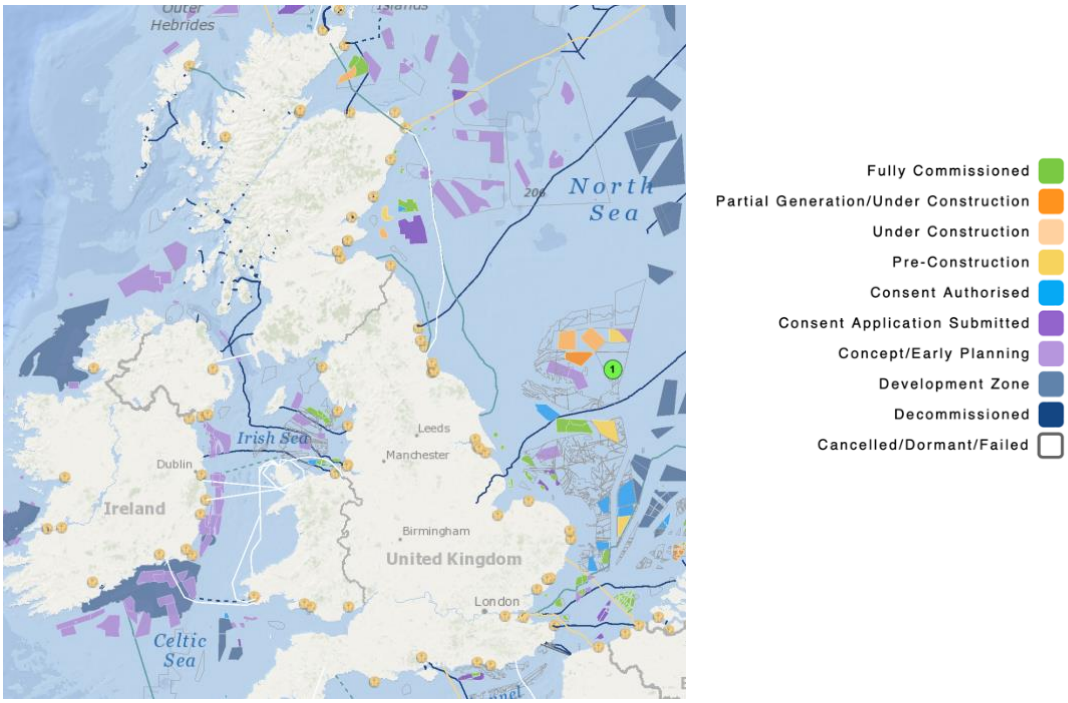
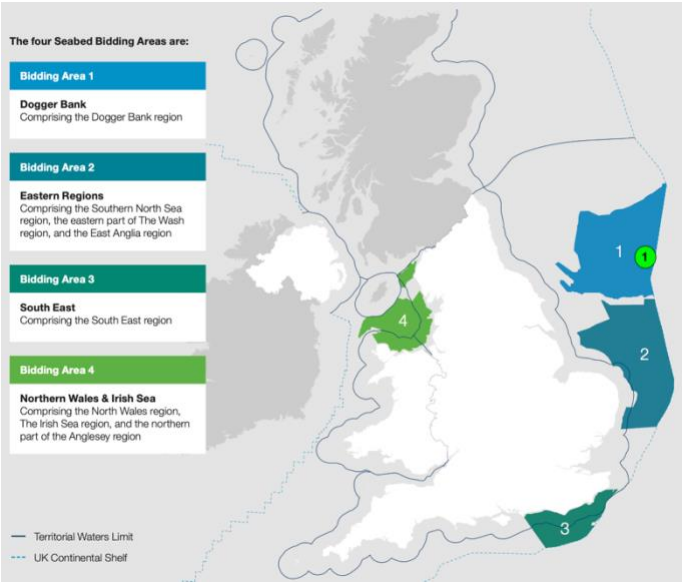


Fig 6: Map of wind farm allocated zones, as well as wind farm developments at different stages [40].



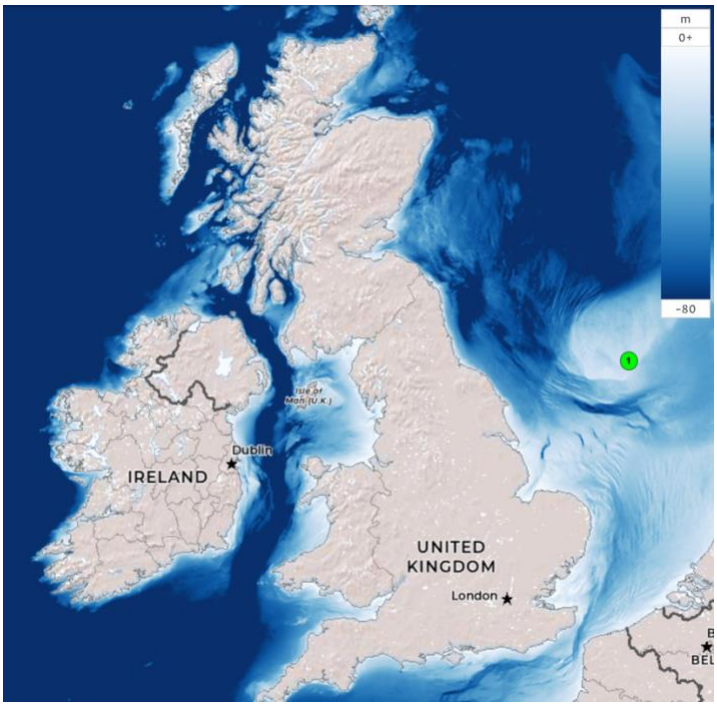


Fig 7 (Left) and Fig 8 (Right): Seabed bidding areas allocated by Crown Estate for round 4 (Left) and water depth chart around UK economic zone (Right) [41][42].

In figure 7, the fourth round of seabed bidding areas allocated by the Crown Estate can be seen [8]. Bidding is also available in area 3 near the channel, and in the Irish Sea. Upon reference to the depth chart in figure 8, it would appear that the south-eastern portion of area 1 is relatively shallow with a mean depth of 20m, and is roughly 120km from the shore. Although area 3 near the Irish Sea appears also relatively shallow and is much closer to shore, the space is cluttered with existing wind developments, and may result in additional restrictions due to close proximity with other farms and also maritime traffic.

As such, the green map reference 1 will be chosen as the offshore development location, offering high mean wind speeds of 9.91m/s measured at a height of 100m, a 120km distance to shore, and has ample separation of roughly 20km from neighbouring development projects [35]. The precise location is 54.636°, 2.461°, and known as Dogger Bank. With an average depth of 20m, fixed-bottom turbines can be utilised, offering lower installation and maintenance costs as opposed to more complex floating platforms [43]. Considering the prime location available, a power generation goal of 350 GWh/year will be chosen for this question.

3.2 Wind turbine choice

Wind turbines are classified with the International Electrotechnical Commission (IEC) system from class I to class III, ranging from a rated average wind speed of 10, 8.5, 7.5m/s for classes I, II and III respectively. An additional classing from A to C refers to the rated turbulence intensity of wind, from 16% at class A, 14% at B and 12% at C [44]. From [35], it can be seen that at the location, the mean wind speed is 9.91m/s at 100m, 10.31m/s at 150m and 10.53m/s at 200m. In addition, a 2023 study recorded 90th-percentile turbulence intensities of not more than 8% at and above 10m/s speeds at Dogger Bank East, suggesting a class IC turbine is required [45].

The world's four most powerful offshore turbines are listed in the table below, including power output, key characteristics and current deployments on wind farms. The largest-possible turbines are considered as they are cheaper per unit power output, and space/visual impact is far less of a limiting factor when considering offshore wind farms.

Manufacturer	Model	Power Rating (MW)	Rotor Diameter (m)	Turbine Height (m)	Blade length (m)	Swept Area (m^2)	W/m^2	Reported Annual Energy	Capacity Factor	IEC Class	Frequency	Serial Production

								Production AEP (GWh)				
Windey (CN)	WD245-15000	15	245	Site-specific	120	47,144	318	n/a	n/a	IC	50/60Hz	2023 [46]
Vestas (DK)	V236-15.0 MW	15	236	Site-specific	115.5	43,724	343	80	60%	IC	50/60Hz	2023 [47]
Siemens Gamesa (ES)	SG 236 DD	14/15	236	Site-specific	115	43,500	320	n/a	n/a	IC	50/60Hz	2024 [48]
GE Renewables (FR)	Haliade-X 14 MW	14	220	138-150m	107	38,013	387	74	61%	IC	50/60Hz	2023 [49]

Table 2: Table with characteristics of highest-output commercially-available wind turbines as of report date [46][47][48][49].

Seeing as no power-wind velocity curves are available for any of the models, due to more complete information being available on the Haliade-X 14 MW turbine in addition to a smaller physical footprint, it will be used as the turbine of choice. A power-wind velocity curve was then generated using the model developed by Saint-Drenan et. al [50], using the turbine parameters from the above table as well as cut-in wind speed of 3m/s and cut-out wind speed of 34m/s [51]. The tallest hub height of 150m was selected to maximise power output. Seeing as the region is remote, the visual impact is negligible and the increased cost may be somewhat compensated by cheaper installation due to shallow waters.

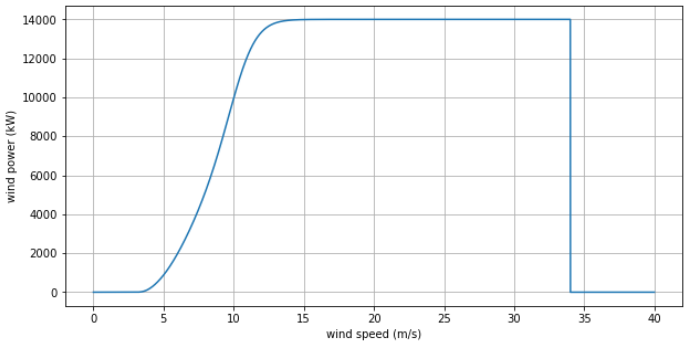


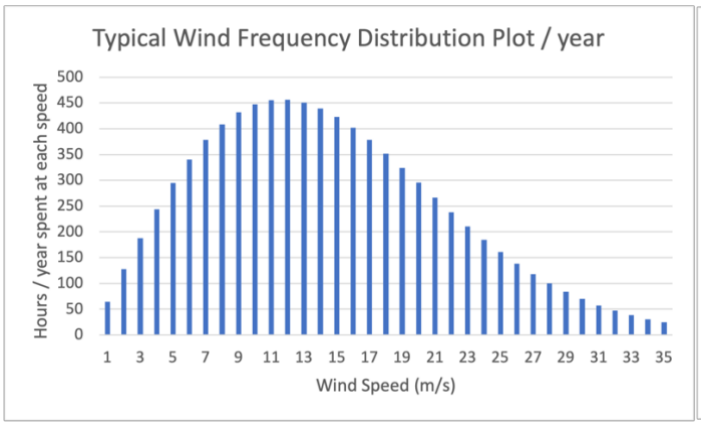
Fig 9: Power versus mean wind speed characteristic curve for GE Haliade-X 14 MW [49][50].

3.3 Energy production analysis

Using this value will also skip the necessity of correcting for hub height wind speeds as wind speeds at 150m are already available via [35]. However, this can be done using the following equation [52]:

$$V(\mathrm{h}_2) = V(\mathrm{h}_1)\frac{\ln\left(\frac{\mathrm{h}_2 - d}{z_0}\right)}{\ln\left(\frac{(\mathrm{h}_1 - d)}{z_0}\right)}$$

Where d is 0-70% of nearby obstacle height, z_0 is the roughness length equal to 0.0002 for sea, h_2 is the new height and h_2 is measured height. Using the Rayleigh distribution and a scale factor of $\sigma = \frac{2\bar{V}}{\sqrt{\pi}}$, a value known to describe wind speed accurately, a typical wind frequency distribution plot was generated for a mean wind speed of 10.31m/s [52]:



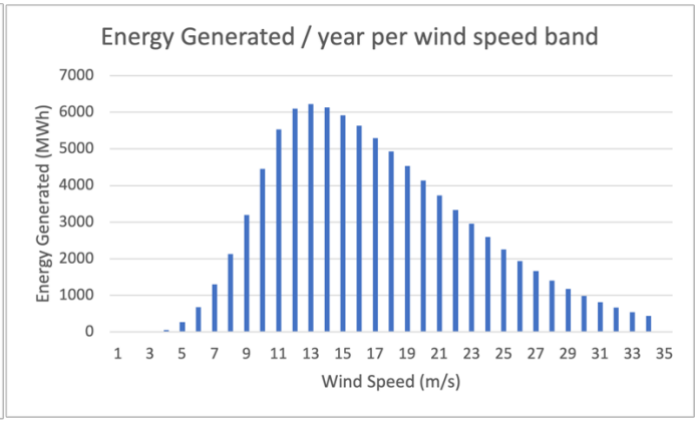


Fig 10 (Left), Fig 11 (Right): Typical wind frequency distribution plot per year (Left) and Energy generated from each wind speed band per year (Right.

The values in figure 10 are then multiplied by the corresponding Power curve wattage, resulting in the power generated per year from each wind speed, in MWh seen in figure 11 above. These values are summed to give a net production of 90.91 GWH/year, resulting in a capacity factor of 70.6% for a maximum generating capacity of 14.7 MW. Using this value, only 4 turbines are needed to reach the 350 GWh/year goal. However, systems are not expected to operate at specified performance all year round due to maintenance and repairs needed. Typical contractual guarantees are 97% availability for onshore wind and 95% for offshore [53]. This would reduce the generated power to 86.36 GWH/year and a capacity factor of 67.0%. Required AC-DC conversion from wind turbine generators typically incurs losses of 2%, and transmission losses to the grid, following a AC-DC conversion, linkage to interlink, and transmission from interlink to shoreline typically incurs an additional 3.5% loss [54][55]. This brings the expected annual energy production (AEP) to 81.61 GWh/year, with a capacity factor of 63.4%. Upon considering additional factors such as wake loss, the power generation may be closer to the rated values seen in table 1. From these calculations, the production goal of 350 GWh/year can be met with 5 Haliade-X 14 MW turbines, which will produce an estimated 408.05 GWh/year.

3.4 Network topology

The radial network topology will be used, providing ease of installation and maintenance at the cost of lack of redundancy [52]. This is deemed acceptable as the wind farm is relatively small scale when compared to the neighbouring Dogger Bank farm phases producing 4.8 GW [56]. As such, the significant increase in cabling costs required for single-sided ring topology is not required. A wind power rose for the location at a height of 150m was extracted from [35], seen in figure 12 below. This suggests the turbine layout should be perpendicular to the southwest direction, with each turbine also facing the southwest direction. This would minimise the width required for the farm as successive turbines will be perpendicular to the wind direction, avoiding wake losses. A typical separation of 4 rotor diameters, or 880m will be used between each turbine [57], resulting in a net farm width of 3,520m.

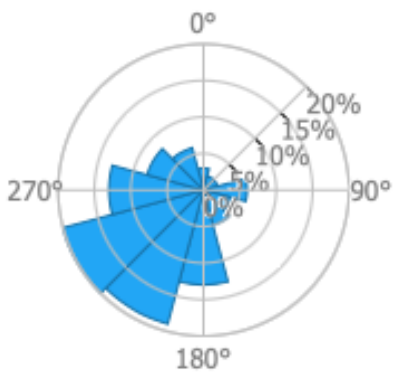


Fig 12: Wind Power Rose for location from [35]

3.5 Steps taken for Intermittency of Wind Energy

From figure 13 below, it can be seen that on average at the location, wind varies by 3% hourly, up to 20% monthly and up to 10% across the year [35]. Wind power is known to be a variable resource and thus mitigation factors should be in place if it is to continue as the predominant source of renewable energy.

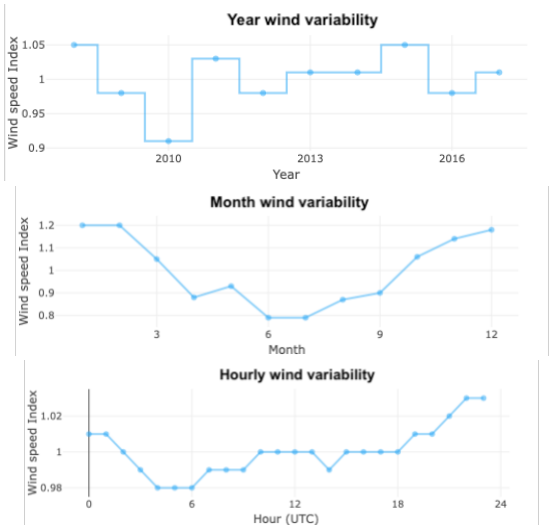


Fig 13: Wind variability across hourly, monthly and yearly timespans for site location [35].

Studies have suggested a 15-22% installed intermittent capacity is necessary, but National Grid has declared that the current distribution grid can manage on-off wind energies with the implementation of a "smart grid" system, expanding on energy storage methods such as pumped water hydroelectricity as well as increasing interconnection links with the rest of Europe [59]. In 2016, 10 coal and gas-powered generators were contracted by the National Grid to inventory spare capacity for all generation modes. In addition, technologies like pumped-storage hydroelectricity and grid-scale battery storage is being developed, adding to 3.8 GW of storage capacity as of 2021 [60]. Thus, as reflected by the priorities of the National Grid, combatting wind variability can be done through diversification of generation methods, including temporarily reverting to fossil fuels if necessary. An additional buffer can be made with advancements in storage technologies such as battery and PSH storage.

3.6 Assessment of Environmental Impact

It has been reported that wind turbine foundations provide favourable habitats for marine life due to improved feeding conditions from habitat alteration. The result is attracting fish species favouring rocky and sand-bottom habitats, intruding on the local cycle and habitats of marine wildlife. Through this mechanism, fixed offshore wind turbines have been reported to contribute to nutrient concentration through localised vertical mixing. Through changing current velocities and facilitating the growth of biomass, the consumption of oxygen and degradation of biomass will increase, intensifying eutrophication [61]. However, further study is required on this topic to develop a comprehensive understanding.

Wind farm cables are also known to generate small magnetic fields within the range of 1-100 μV/cm, similar to those of some marine prey. This can interfere with prey finding of some species. In addition, larger turbines like the Haliade-X chosen can generate larger fields which may also interfere with migration and orientation capabilities of larger predators, a study finds [61]. Offshore turbines can also generate significant underwater noise, during installation and operation. A monitoring program was carried out in 2018 at the Scroby Sands wind farm in the UK, and discovered a significant decrease in harbour seal count, whereas the number of grey seals increased. An additional study in 2020 found a significant decrease in number of fish within 140m of each turbine [61].

In addition, corrosion on turbines can lead to emissions of eco-toxicological material like Aluminium, Zinc and Indium [61]. Additional research is still required to determine the long-term effects of such materials. Additional mechanisms such as leaching, weathering or material loss can also result in release of organic compounds, solvents, binding agents, pigments and fillers into the seawater from the coatings of turbine parts. Their environmental effects are still pending further study [61].

3.7 Effect of Global Warming on Offshore Wind

A 2024 study modelled the increased occurrence of extreme high or low weather events through 2021-2040 and 2061-2080 across multiple sea regions of the UK [62]. Due to more extreme weather, low speed wind events are predicted to occur more frequently from 2061-2080 below the turbine cut-in speeds of 3m/s, and are predicted to be especially low across summer. The effect is more prominent for regions near the coastline, and less prominent in the North and Southern regions of the UK specifically. The modelling was done using the UKCP18 daily mean, and at a similar hub height of 151m.

3.8 Cost estimate for 15-year lifespan of system

The area required for the farm is a small portion of the Dogger Bank bidding area in the round 4 allocations from the Crown Estate [40]. In terms of area required, a separation of 4x rotor diameter is used perpendicular to the wind direction, and 8x rotor diameter parallel, resulting in an area of 3520x1760m = 6,195,200m^2. The company RWE was granted a lease on a Dogger Bank location for a 3000 MW wind farm during leasing round 4 by the Crown Estate [63]. Although the duration is not disclosed, the price was set at £82,552 per MW per year, which is stated as the lowest price award during the auction. Following the same pricing, the question proposes a 70 MW farm seabed lease for a duration of 15 years, costing a total of £86.68m.

The GE Haliade-X 14 MW costs between £8.7-10.3m per unit, suggesting an initial purchase cost between £43.5-51.5m [64]. Seeing as the higher hub-height specification was chosen, the upper band cost of £51.5m is considered. A study by Rystad Energy in 2020 estimated that foundation costs range between £2.4-3.2m [65]. Seeing as this is for foundations ranging between 10-69m, for an average depth of 20m at Dogger Bank, a proportionally-accurate £2.53m is considered per foundation, amounting to £12.63m. Installation costs were reported at £1.66m per turbine, including the turbine itself and foundation, costing £8.3m. Thus, initial expenditure on turbines and foundations adds to £72.43m.

A statistical analysis suggested that, for a 15-year project, the cost of turbines and installation made up 42.3% of a typical project budget, excluding the cost of land acquisition [66]. The remaining budget is made up of development and project management at 3.5%, cabling at 5%, offshore substation purchase at 3.5%, maintenance costs at 18.9%, operations costs at 9.3%, installation of substation, cabling and others at 15.4%, and finally decommissioning at 1.8%. As such, the cost can be estimated at £171.23m for the farm life cycle, and £86.68m for land acquisition across 15 years, totalling £257.91m. With an expected net production of 6,120.75 GWh across its lifecycle, this will produce an average electricity price of £42.17 per MWh, 14% cheaper than the average contracted selling price of £48 per MWh for upcoming projects due to start by 2026/27 [67].

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